

Chapter 2

Bohr Model of the Atom

Relevant sections of Foot for this topic include §1.3 and §1.4 [1].

The Bohr model of a one-electron atom is a classical mechanics description of an inherently quantum system – and electron orbiting a charged nucleus. The slight-of-hand involved is that one imposes *quantization conditions* on the classical model that make *some* sense if you already know some of the main results of quantum mechanics, but these must essentially be taken as guesses whose validity stands only on the resulting model’s ability to accurately predict experimental outcomes. This process of adding some quantization to classical models does not generalize very well to other quantum systems, but works *astonishingly* well for one-electron atoms, as we will see below.

2.1 Classical mechanics

We begin with the problem of solving for the motion of a spinless electron of charge $-e$ and mass m in the presence of a (for the moment, fixed) structureless nucleus of charge $+Ze$. The Coulomb force attracting the electron to the nucleus is given by

$$\mathbf{F} = \frac{(Ze)(-e)}{4\pi\epsilon_0} \frac{1}{r^2} \hat{\mathbf{e}}_r \quad (2.1)$$

$$= -\frac{Ze^2}{4\pi\epsilon_0} \frac{1}{r^2} \hat{\mathbf{e}}_r \quad (2.2)$$

where $\hat{\mathbf{e}}_r$ points from the nucleus (set at the center of a spherical coordinate system) to the electron’s position.

We can start by equating the electrostatic force with mass times the radial acceleration,

$$F = ma$$

$$-\frac{Ze^2}{4\pi\epsilon_0 r^2} = m(\ddot{r} - r\omega^2)$$

where $\omega = v/r$ is the angular frequency of the electron's orbit. In particular, we will confine our description to a case where the electron moves in a circular orbit with fixed radius r , in which case $\ddot{r} = 0$ and we have

$$\frac{Ze^2}{4\pi\epsilon_0 r^2} = m\frac{v^2}{r} \quad (2.3)$$

A circular orbit with radius r and tangential speed v will complete a cycle in time T such that

$$vT = 2\pi r$$

$$T = \frac{2\pi r}{v}$$

therefore

$$\omega = \frac{2\pi}{T} = \frac{v}{r}. \quad (2.4)$$

Combining this with Eq. (2.3) (divided by mr) gives

$$\frac{Ze^2}{4\pi\epsilon_0 mr^3} = \omega^2. \quad (2.5)$$

Next, we can calculate the energy. First, the kinetic energy is given by

$$K = \frac{1}{2}mv^2$$

$$= \frac{1}{2}m\omega^2 r^2$$

$$= \frac{Ze^2}{8\pi\epsilon_0 r}. \quad (2.6)$$

The potential energy is straight from electrostatics,

$$U = -\frac{Ze^2}{4\pi\epsilon_0 r}, \quad (2.7)$$

so the total energy comes out to

$$\begin{aligned}
E &= K + U \\
&= \frac{1}{2} \frac{Ze^2}{4\pi\epsilon_0 r} - \frac{Ze^2}{4\pi\epsilon_0 r} \\
&= -\frac{Ze^2}{8\pi\epsilon_0 r}.
\end{aligned} \tag{2.8}$$

2.2 de Broglie Quantization

Next, we will introduce, essentially *ad hoc*, a quantization condition. Foot [1] does this by assuming that the angular momentum is quantized, so we will take a different approach here just to illustrate another path to the same conclusion.

Following the proposal that Louis de Broglie made in his doctoral thesis in 1924, we will assume that the electron has a wavelength associated with it (λ_{dB}), such that

$$p = \hbar k_{\text{dB}} \tag{2.9}$$

where $k_{\text{dB}} \equiv 2\pi/\lambda_{\text{dB}}$. Put another way, we have

$$\lambda_{\text{dB}} = 2\pi \frac{\hbar}{p}. \tag{2.10}$$

Given this assumption, we further postulate that the stable orbits of the hydrogen atom will correspond to orbits that satisfy *periodic boundary conditions* (which is to say orbits such that the phase of this wave after one full orbit is the same as its initial phase). So the circumference of the orbit must be an integer times the (de Broglie) wavelength:

$$\begin{aligned}
2\pi r_n &= n\lambda_{\text{dB}} \quad \text{such that } n \in \mathbb{N} \\
&= 2\pi \frac{\hbar}{mv_n} n
\end{aligned} \tag{2.11}$$

and, dividing both sides by 2π , we find that

$$\begin{aligned}
r_n &= \frac{\hbar}{mv_n} n \\
&= \frac{\hbar}{m\omega_n r_n} n
\end{aligned} \tag{2.12}$$

where I have used (2.4). Rearranging and squaring both sides gives us

$$\omega_n^2 r_n^4 = \frac{\hbar^2}{m^2} n^2. \quad (2.13)$$

Substitution of (2.5) gives

$$\begin{aligned} \frac{Ze^2}{4\pi\epsilon_0} \frac{r_n}{m} &= \frac{\hbar^2}{m^2} n^2 \\ \frac{Ze^2}{4\pi\epsilon_0} r_n &= \frac{\hbar^2}{m} n^2, \end{aligned}$$

but we know from (2.8) that

$$r_n = -\frac{Ze^2/4\pi\epsilon_0}{2E_n},$$

so

$$-\frac{1}{2} \left(\frac{Ze^2}{4\pi\epsilon_0} \right)^2 \frac{1}{E_n} = \frac{\hbar^2}{m} n^2$$

and therefore

$$\begin{aligned} E_n &= -\frac{m}{2\hbar^2} \left(\frac{Ze^2}{4\pi\epsilon_0} \right)^2 \frac{1}{n^2} \\ &= -hcZ^2 R_\infty \frac{1}{n^2}, \end{aligned} \quad (2.14)$$

where

$$R_\infty \equiv \frac{me^4}{8c\epsilon_0^2 h^3} \quad (2.15)$$

is the *Rydberg constant*. Slightly more practically useful than this is the *Rydberg frequency*,

$$cR_\infty = 3.289\,841\,960\,2508(64) \times 10^{15} \text{ Hz}.$$

The constant R_∞ is usually what is meant when people refer to the “Rydberg constant,” though as we will see below, there is some ambiguity about this as R_∞ does not include the effect of finite nuclear mass and some folks really mean the mass-corrected constant.

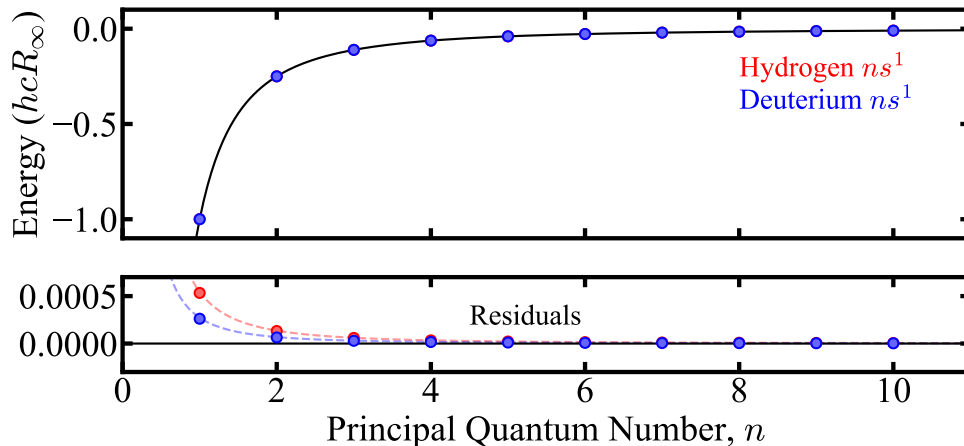


Figure 2.1: Empirical values for the energies of the first ten ns^1 states of hydrogen (red) and deuterium (blue) with the predictions of the R_∞ Bohr model (black). The bottom panel shows the *residuals*, or the differences between the data and the model, with the finite-nuclear-mass-corrected Bohr model shown as dashed lines.

Equation (2.14) is the famous Bohr formula for the energies of the states of the hydrogen atom, and the integer n is known as the *principal quantum number*. Transitions in this model occur between states with different principal quantum numbers, and when these transitions occur optically (as single-photon transitions), they correspond to photons with energies given by the difference between the two level energies.

Since the hydrogen atom is the simplest atom to treat theoretically, considerable effort has been dedicated to measuring its spectrum with increasing precision. The Bohr model of Eq. (2.14) is plotted along with empirical values for the energies of the ns^1 states in Figure 2.1. Considering that this model comes from *ad hoc* quantization and classical mechanics, the agreement with experimental values for a system that is known to be highly affected by quantum mechanics is frankly remarkable.

Another question we can answer with this model is the *size of the atom*, which is to say how large is the orbit of the electron about the (fixed) proton?

Plugging (2.14) back into (2.2) gives

$$\begin{aligned}
r_n &= -\frac{Ze^2}{8\pi\epsilon_0} \frac{1}{E_n} \\
&= \frac{Ze^2}{8\pi\epsilon_0} \frac{\hbar^2}{m} \frac{8\pi\epsilon_0}{Ze^2} \frac{4\pi\epsilon_0}{Ze^2} n^2 \\
&= \frac{\hbar^2}{\left(\frac{Ze^2}{4\pi\epsilon_0}\right) m} n^2
\end{aligned} \tag{2.16}$$

$$\equiv \frac{a_0}{Z} n^2. \tag{2.17}$$

The length,

$$a_0 \equiv \frac{\hbar^2}{\left(\frac{e^2}{4\pi\epsilon_0}\right) m} \tag{2.18}$$

is known as the *Bohr radius* and is about half of an angstrom ($a_0 = 5.291\,772\,109\,03(80) \times 10^{-11}$ m, and $1 \text{ \AA} \equiv 10 \times 10^{-11}$ m). For comparison, the radius of the proton is less than $1 \text{ fm} = 10^{-15}$ m. Expressed as a volume ratio, the atom occupies roughly 10^{14} times the volume of the proton.

We can also ask, what is the electron's angular momentum¹ in these Bohr orbits? Using basic classical mechanics, we have

$$\begin{aligned}
L_n &= m r_n v_n \\
&= m \omega_n r_n^2 \\
&= m \frac{n\hbar}{m r_n^2} r_n^2 \\
&= n\hbar.
\end{aligned} \tag{2.19}$$

So the classical angular momentum in this circular-orbit model is quantized in units of $\hbar$, though you will likely recall that quantum mechanics allows states with zero angular momentum. The Bohr model, then, seems to correspond most-closely to what we would call “stretched” states in the quantum model, which is to say the states with the maximum eigenvalues for J^2 and J_z .

¹We will in this section assume the electron has no spin, so all of its angular momentum is orbital angular momentum.

2.3 Finite Nuclear Mass

The Rydberg constant R_∞ was derived assuming the mass of the nucleus is infinite compared to the mass of the electron. As such, this model would predict that the spectrum of hydrogen would look identical to the spectrum of its heavier sibling, deuterium, whose nucleus consists not only of a proton but also a neutron, and is therefore roughly twice as heavy. The differences between the spectra of hydrogen (H) and deuterium (D) are not visible on the gross structure shown in the upper panel of Fig. 2.1, but do appear in the residuals (lower panel of Fig. 2.1). As expected, the “ R_∞ Bohr model” of Eq. (2.14) that assumes an infinitely heavy nucleus matches the heavier nucleus (D) better than the light one (H).

As it turns out, it is not too difficult to take the finite nuclear mass into account for one-electron atoms. We simply replace the electron mass with the *reduced* electron mass,

$$\tilde{m} \equiv \frac{mM_{\text{nuke}}}{m + M_{\text{nuke}}}. \quad (2.20)$$

Since the Rydberg constant is proportional to m , it can be corrected to account for this by scaling it:

$$\tilde{R} = R_\infty \frac{\tilde{m}}{m} = R_\infty \frac{M_{\text{nuke}}}{m + M_{\text{nuke}}}. \quad (2.21)$$

The finite-nuclear-mass-corrected Bohr model is shown for hydrogen (red, dashed) and deuterium (blue, dashed) in the bottom panel of Fig. 2.1, and agrees with the empirical values to better than 4 digits. The particular difference we have calculated is a type of *isotope shift* called the *mass shift*. Later, we will see that there are other types of isotope-dependent changes to atomic spectra that arise from things like the finite size of the nucleus (yes, the electron really does go *inside* the nucleus, even inside the proton itself!) and its electromagnetic multipole moments.

2.4 Scale of Relativistic Corrections

We have already seen how useful the Bohr model is for estimating basic properties of hydrogen; since it’s a classical model, it’s super easy to just

calculate the orbit size, or the angular momentum, or the corrections from finite nuclear mass.

Another aspect of the system worth examining is the (potentially relativistic) speed of the electron. Taking $v_n = \omega_n r_n$ for circular motion and plugging this into (2.13) we have

$$\begin{aligned} v_n &= \omega_n r_n \\ &= \frac{1}{r_n} \omega_n r_n^2 \\ &= \frac{1}{r_n} \frac{\hbar}{m} n. \end{aligned} \tag{2.22}$$

Equation (2.16) tells us what the radius is,

$$r_n = \frac{4\pi\epsilon_0 \hbar^2}{Ze^2 m} n^2 \tag{2.23}$$

and we get

$$\begin{aligned} v_n &= \frac{Ze^2}{4\pi\epsilon_0 \hbar} \frac{1}{n} \\ &= c \times \left(\frac{Ze^2}{4\pi\epsilon_0 c \hbar} \right) \frac{1}{n} \\ &= c \times Z\alpha \frac{1}{n}. \end{aligned} \tag{2.24}$$

Here, we have noticed that there is a dimensionless quantity we've been carrying around in many of these derivations and it's known as the *fine structure constant*:

$$\alpha \equiv \frac{e^2}{4\pi\epsilon_0 \hbar c} \approx \frac{1}{137}. \tag{2.25}$$

The fine structure constant has uses beyond atomic calculations, particularly in quantum electrodynamics (QED) where it specifies the strength of the electromagnetic interaction for perturbation theory, but here it is mostly just a convenient way to gather terms.

Given that the fine structure constant is of order $\alpha \approx 1\%$ and we calculated the electron's speed for the hydrogen atom to be $v_n = c\alpha/n$, we see that the electron is moving at an appreciable fraction of the speed of light,

even in the lightest atom. In fact, had we allowed for higher nuclear charge $Z > 1$, we'd have $v_n = c \times Z\alpha/n$.²

Other useful expressions can be likewise obtained in terms of α as follows. The Bohr radius and energy are

$$a_0 = \frac{\hbar}{mc} \frac{1}{\alpha} \quad (2.26)$$

and

$$E_n = -\frac{1}{2}mc^2 Z^2 \alpha^2 \frac{1}{n^2}. \quad (2.27)$$

The electron's translational momentum is

$$p_n = m Z\alpha c/n, \quad (2.28)$$

the orbital frequency is

$$\omega_n = Z^2 \frac{mc^2}{\hbar} \alpha^2 \frac{1}{n^3}, \quad (2.29)$$

and the angular momentum is, as we observed earlier,

$$l_n = mv_n r_n = n\hbar. \quad (2.30)$$

Having established that the electron's motion is relativistic, we need to determine how large of a correction this will require to the Bohr formula. The relativistic expression for the electron's kinetic energy is

$$K = (\gamma - 1)mc^2 \quad (2.31)$$

where

$$\gamma \equiv \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}. \quad (2.32)$$

²It is interesting to consider what happens here when $Z \geq 137$, as our simple formula would imply that the electron could move faster than the speed of light. Clearly, this cannot happen, as these equations treat the motion non-relativistically. However, if we consider the binding energy of the ground state of a hydrogenic Bohr atom with $Z = 137$, we will find that it is the same order of magnitude as mc^2 , the energy of the rest mass of the electron. Clearly, something interesting happens as limits of these types are approached, even if we admit that our classical estimates of where this occurs require substantial modification.

Expanding (2.32) in powers of v/c gives

$$K \approx \frac{1}{2}mv^2 \left(1 + \frac{3}{4} \left(\frac{v}{c} \right)^2 \right) \quad (2.33)$$

$$= \frac{1}{2}mv^2 \left(1 + \frac{3Z^2\alpha^2}{4n^2} \right) \quad (2.34)$$

and we see that we can expect the fractional corrections to the energy from relativity to be of order $\approx Z^2\alpha^2/n^2$.